CNC PROGRAM SHEET

Part: MOTOR MOUNT | Rev: A | Date: 2026-06-23

Programmer: CNC-AI

Material: ALUMINIUM

Max RPM: 10000 Coolant: through spindle

Setups: 12

Part envelope: 55.3 x 104.8 x 21.5 mm

COORDINATE SYSTEM & WORK ZERO (G54)

Work zero convention	--	**Part height (Y)**	-- mm
CAD up-axis	--	**X range (machine)**	-- to -- mm
Work zero in CAD	X=0 Y=0 Z=0	**Z range (machine)**	-- to -- mm
G54 meaning	Set controller G54 offset so that X0 Y0 Z0 = top face centre of part	**Safe rapid Y**	10 mm above work zero

TOOL LIST

T#	Tool ID	Description	Dia (mm)	Manufacturer	Grade	Used for
T01	EM_2FL_1MM	2-flute solid carbide end mill 1mm â€" aluminium geometry	1.0	Sandvik Coromant	1630	contour_mill
T02	CD_60_2MM	60-degree combined center drill, 2mm tip, 5mm body	2.0	Sandvik Coromant	H10F	spot_drill
T03	EM_2FL_2MM	2-flute solid carbide end mill 2mm â€" aluminium geometry	2.0	Sandvik Coromant	1630	contour_mill
T04	DRILL_2.5MM	Solid carbide jobber drill 2.5mm	2.5	Sandvik Coromant	H10F	twist_drill
T05	DRILL_3.0MM	Solid carbide jobber drill 3.0mm	3.0	Sandvik Coromant	H10F	twist_drill
T06	EM_2FL_3MM	2-flute solid carbide end mill 3mm â€" aluminium geometry	3.0	Sandvik Coromant	1630	contour_mill, counterbore_mill
T07	SPOT_090_4MM	90-degree solid carbide spot drill, 4mm shank	4.0	Sandvik Coromant	H10F	spot_drill
T08	EM_2FL_4MM	2-flute solid carbide end mill 4mm â€" aluminium geometry	4.0	Sandvik Coromant	1630	counterbore_mill
T09	SLOT_4MM	2-flute solid carbide slot mill 4mm â€" short cut length for rigidity	4.0	Sandvik Coromant	1630	slot_mill
T10	DRILL_5.0MM	Solid carbide jobber drill 5.0mm	5.0	Sandvik Coromant	H10F	twist_drill
T11	EM_2FL_6MM	2-flute solid carbide end mill 6mm â€" aluminium geometry	6.0	Sandvik Coromant	1630	contour_mill
T12	CHAM_90_6MM	90-degree solid carbide chamfer mill 6mm, 2-flute	6.0	Sandvik Coromant	1630	chamfer_mill
T13	SLOT_6MM	2-flute solid carbide slot mill 6mm â€" short cut length for rigidity	6.0	Sandvik Coromant	1630	slot_mill
T14	EM_2FL_8MM	2-flute solid carbide end mill 8mm â€" aluminium geometry	8.0	Sandvik Coromant	1630	circular_interp, counterbore_mill
T15	SLOT_8MM	2-flute solid carbide slot mill 8mm â€" short cut length for rigidity	8.0	Sandvik Coromant	1630	slot_mill
T16	EM_2FL_20MM	2-flute solid carbide end mill 20mm â€" aluminium geometry	20.0	Sandvik Coromant	1630	circular_interp
T17	FACEMILL_40MM	40mm indexable face mill â€" 6 inserts	40.0	Sandvik Coromant		face_mill
T18	FACEMILL_63MM	63mm indexable face mill / shell mill â€" 5 inserts	63.0	Sandvik Coromant		face_mill

SETUP 1 -- -Z approach (rear face)

PRINCIPAL | 70 clusters | 128 operations

Fixture	90° angle plate or tombstone fixture.
Description	Rear-face setup. Rotate part 90° so rear face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G54
Origin X/Y/Z	X=CENTER (X=0.000mm) Y=CENTER (Y=-20.711mm) Z=TOP
Face (W x H)	55.3 x 104.8 mm
Max depth	21.5 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL COUNTERBORE	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
2	ALU 2.5 DRILL COUNTERBORE	T04	2.5	0	3.87	10000	380	3.869	--	--	00:01
3	ALU 8 ENDMILL COUNTERBORE RF	T14	8.0	0	3.87	10000	600	3.869	0.1	0.1	00:02
4	ALU 8 ENDMILL COUNTERBORE FINISH	T14	8.0	0	3.87	10000	600	3.869	0.0	0.0	00:02
5	ALU CENTER DRILL COUNTERBORE	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
6	ALU 2.5 DRILL COUNTERBORE	T04	2.5	0	3.87	10000	380	3.868	--	--	00:01
7	ALU 8 ENDMILL COUNTERBORE RF	T14	8.0	0	3.87	10000	600	3.868	0.1	0.1	00:02
8	ALU 8 ENDMILL COUNTERBORE FINISH	T14	8.0	0	3.87	10000	600	3.868	0.0	0.0	00:02
9	ALU CENTER DRILL COUNTERBORE	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
10	ALU 2.5 DRILL COUNTERBORE	T04	2.5	0	4.42	10000	380	4.419	--	--	00:01
11	ALU 3 ENDMILL COUNTERBORE RF	T06	3.0	0	4.42	10000	300	3.0	0.1	0.1	00:05
12	ALU 3 ENDMILL COUNTERBORE FINISH	T06	3.0	0	4.42	10000	300	3.0	0.0	0.0	00:03
13	ALU 4 ENDMILL COUNTERBORE RF	T08	4.0	0	4.42	10000	360	4.0	0.1	0.1	00:04
14	ALU 4 ENDMILL COUNTERBORE FINISH	T08	4.0	0	4.42	10000	360	4.0	0.0	0.0	00:03
15	ALU CENTER DRILL COUNTERBORE	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
16	ALU 2.5 DRILL COUNTERBORE	T04	2.5	0	4.42	10000	380	4.419	--	--	00:01
17	ALU 3 ENDMILL COUNTERBORE RF	T06	3.0	0	4.42	10000	300	3.0	0.1	0.1	00:05
18	ALU 3 ENDMILL COUNTERBORE FINISH	T06	3.0	0	4.42	10000	300	3.0	0.0	0.0	00:03
19	ALU 4 ENDMILL COUNTERBORE RF	T08	4.0	0	4.42	10000	360	4.0	0.1	0.1	00:04
20	ALU 4 ENDMILL COUNTERBORE FINISH	T08	4.0	0	4.42	10000	360	4.0	0.0	0.0	00:03
21	ALU CENTER DRILL COUNTERBORE	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
22	ALU 2.5 DRILL COUNTERBORE	T04	2.5	0	4.42	10000	380	4.419	--	--	00:01
23	ALU 3 ENDMILL COUNTERBORE RF	T06	3.0	0	4.42	10000	300	3.0	0.1	0.1	00:05
24	ALU 3 ENDMILL COUNTERBORE FINISH	T06	3.0	0	4.42	10000	300	3.0	0.0	0.0	00:03
25	ALU 4 ENDMILL COUNTERBORE RF	T08	4.0	0	4.42	10000	360	4.0	0.1	0.1	00:04
26	ALU 4 ENDMILL COUNTERBORE FINISH	T08	4.0	0	4.42	10000	360	4.0	0.0	0.0	00:03
27	ALU CENTER DRILL COUNTERBORE	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
28	ALU 2.5 DRILL COUNTERBORE	T04	2.5	0	4.42	10000	380	4.419	--	--	00:01
29	ALU 3 ENDMILL COUNTERBORE RF	T06	3.0	0	4.42	10000	300	3.0	0.1	0.1	00:05
30	ALU 3 ENDMILL COUNTERBORE FINISH	T06	3.0	0	4.42	10000	300	3.0	0.0	0.0	00:03
31	ALU 4 ENDMILL COUNTERBORE RF	T08	4.0	0	4.42	10000	360	4.0	0.1	0.1	00:04
32	ALU 4 ENDMILL COUNTERBORE FINISH	T08	4.0	0	4.42	10000	360	4.0	0.0	0.0	00:03
33	ALU 63 FACEMILL BORE RF	T18	63.0	0	2.72	4040	3636	1.0	0.0	0.2	00:02
34	ALU 63 FACEMILL BORE FINISH	T18	63.0	0	2.72	4040	3636	0.5	0.0	0.0	00:03
35	ALU 20 ENDMILL BORE	T16	20.0	0	2.72	7960	955	2.724	--	--	00:10
36	ALU CENTER DRILL	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
37	ALU 3 DRILL	T05	3.0	0	1.5	10000	450	1.5	--	--	00:01
38	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.38	10000	300	0.375	0.1	0.1	00:04
39	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.38	10000	300	0.375	0.0	0.0	00:04
40	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.38	10000	300	0.375	0.1	0.1	00:04
41	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.38	10000	300	0.375	0.0	0.0	00:04
42	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.81	10000	300	0.811	0.1	0.1	00:03
43	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.81	10000	300	0.811	0.0	0.0	00:03
44	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.81	10000	300	0.811	0.1	0.1	00:03
45	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.81	10000	300	0.811	0.0	0.0	00:03
46	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.38	10000	300	0.375	0.1	0.1	00:04
47	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.38	10000	300	0.375	0.0	0.0	00:04
48	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.38	10000	300	0.375	0.1	0.1	00:04
49	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.38	10000	300	0.375	0.0	0.0	00:04
50	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.62	10000	300	0.625	0.1	0.1	00:03
51	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.62	10000	300	0.625	0.0	0.0	00:03
52	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.88	10000	300	0.875	0.1	0.1	00:03
53	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.88	10000	300	0.875	0.0	0.0	00:03
54	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	0.38	10000	200	0.375	--	--	00:10
55	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	0.33	10000	200	0.332	--	--	00:10
56	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	0.38	10000	200	0.375	--	--	00:10
57	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	1.5	10000	200	1.5	--	--	00:10
58	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	1.5	10000	200	1.5	--	--	00:10
59	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	1.5	10000	200	1.5	--	--	00:10
60	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	2.5	10000	200	2.5	--	--	00:10
61	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	2.5	10000	200	2.5	--	--	00:10
62	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
63	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
64	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
65	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
66	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
67	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
68	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
69	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
70	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
71	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
72	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
73	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
74	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
75	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
76	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
77	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
78	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
79	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
80	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
81	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
82	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
83	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
84	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
85	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
86	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
87	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
88	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
89	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
90	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
91	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
92	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
93	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
94	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
95	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
96	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
97	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
98	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
99	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
100	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
101	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
102	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
103	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
104	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
105	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
106	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
107	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
108	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
109	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
110	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
111	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
112	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
113	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
114	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
115	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
116	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
117	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
118	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
119	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
120	ALU 40 FACEMILL FACE FINISH	T17	40.0	0	2.0	6370	4586	0.5	0.0	0.0	00:04
121	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
122	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
123	ALU 0 ENDMILL POCKET RF	T??	--	0	--	--	--	--	0.1	0.1	00:00
124	ALU 0 ENDMILL POCKET FINISH	T??	--	0	--	--	--	--	0.0	0.0	00:00
125	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	10000	1000	2.0	--	--	00:10
126	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	10000	1000	2.0	--	--	00:10
127	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	10000	1000	2.0	--	--	00:10
128	ALU 6.0 CHAMFER MILL CHAMFER	T12	6.0	0	--	10000	1000	2.0	--	--	00:10

ESTIMATED SETUP TIME: 07:46 (includes tool changes)

SETUP 2 -- +Z approach (front face)

PRINCIPAL | 7 clusters | 12 operations

Fixture	90° angle plate or tombstone fixture.
Description	Front-face setup. Rotate part 90° so front face is accessible to spindle.
Rotation	None -- standard orientation
WCS	G55
Origin X/Y/Z	X=CENTER (X=0.000mm) Y=CENTER (Y=-20.711mm) Z=TOP
Face (W x H)	55.3 x 104.8 mm
Max depth	21.5 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 2 ENDMILL BOSS RF	T03	2.0	0	0.01	10000	200	0.009	0.1	0.1	00:03
2	ALU 2 ENDMILL BOSS FINISH	T03	2.0	0	0.01	10000	200	0.009	0.0	0.0	00:03
3	ALU 2 ENDMILL BOSS RF	T03	2.0	0	0.02	10000	200	0.019	0.1	0.1	00:03
4	ALU 2 ENDMILL BOSS FINISH	T03	2.0	0	0.02	10000	200	0.019	0.0	0.0	00:03
5	ALU 1 ENDMILL BOSS RF	T01	1.0	0	0.03	10000	100	0.025	0.1	0.1	00:03
6	ALU 1 ENDMILL BOSS FINISH	T01	1.0	0	0.03	10000	100	0.025	0.0	0.0	00:03
7	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.62	10000	300	0.625	0.1	0.1	00:03
8	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.62	10000	300	0.625	0.0	0.0	00:03
9	ALU 3 ENDMILL BOSS RF	T06	3.0	0	0.88	10000	300	0.875	0.1	0.1	00:03
10	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	0.88	10000	300	0.875	0.0	0.0	00:03
11	ALU 4.0 SLOT MILL SLOT	T09	4.0	0	0.41	10000	200	0.408	--	--	00:10
12	ALU 8.0 SLOT MILL SLOT	T15	8.0	0	0.69	10000	400	0.688	--	--	00:10

ESTIMATED SETUP TIME: 01:39 (includes tool changes)

SETUP 3 -- +Y approach (top — default VMC position)

PRINCIPAL | 6 clusters | 10 operations

Fixture	Standard vise or fixture plate, jaws on side faces.
Description	Standard top-face setup. Clamp part with top face accessible to spindle. Machine all features in this setup before repositioning.
Rotation	None -- standard orientation
WCS	G56
Origin X/Y/Z	X=CENTER (X=0.000mm, 27.67mm from xmin edge) Y=CENTER (Z=2.750mm, 10.76mm from zmin edge) Z=TOP
Face (W x H)	55.3 x 21.5 mm
Max depth	104.8 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 8 ENDMILL	T14	8.0	0	33.87	10000	600	8.0	--	--	00:10
2	ALU 1 ENDMILL BOSS RF	T01	1.0	0	6.57	10000	100	1.0	0.1	0.1	00:15
3	ALU 1 ENDMILL BOSS FINISH	T01	1.0	0	6.57	10000	100	1.0	0.0	0.0	00:03
4	ALU 1 ENDMILL BOSS RF	T01	1.0	0	4.27	10000	100	1.0	0.1	0.1	00:11
5	ALU 1 ENDMILL BOSS FINISH	T01	1.0	0	4.27	10000	100	1.0	0.0	0.0	00:03
6	ALU 3 ENDMILL BOSS RF	T06	3.0	0	9.62	10000	300	3.0	0.1	0.1	00:09
7	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	9.62	10000	300	3.0	0.0	0.0	00:03
8	ALU 6 ENDMILL BOSS RF	T11	6.0	0	9.62	10000	500	6.0	0.1	0.1	00:06
9	ALU 6 ENDMILL BOSS FINISH	T11	6.0	0	9.62	10000	500	6.0	0.0	0.0	00:04
10	ALU 6.0 SLOT MILL SLOT	T13	6.0	0	9.99	10000	300	6.0	--	--	00:10

ESTIMATED SETUP TIME: 01:58 (includes tool changes)

SETUP 4 -- -Y approach (bottom — part flipped upside down)

PRINCIPAL | 5 clusters | 9 operations

Fixture	Step-jaw vise or fixture plate with bosses as datums.
Description	Flipped setup — part inverted so bottom face is accessible to spindle. Clamp on previously machined top features or use step-jaw vise.
Rotation	None -- standard orientation
WCS	G57
Origin X/Y/Z	X=CENTER (X=0.000mm, 27.67mm from xmin edge) Y=CENTER (Z=2.750mm, 10.76mm from zmin edge) Z=TOP
Face (W x H)	55.3 x 21.5 mm
Max depth	104.8 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU 1 ENDMILL BOSS RF	T01	1.0	0	6.74	10000	100	1.0	0.1	0.1	00:15
2	ALU 1 ENDMILL BOSS FINISH	T01	1.0	0	6.74	10000	100	1.0	0.0	0.0	00:03
3	ALU 1 ENDMILL BOSS RF	T01	1.0	0	4.1	10000	100	1.0	0.1	0.1	00:11
4	ALU 1 ENDMILL BOSS FINISH	T01	1.0	0	4.1	10000	100	1.0	0.0	0.0	00:03
5	ALU 3 ENDMILL BOSS RF	T06	3.0	0	9.62	10000	300	3.0	0.1	0.1	00:09
6	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	9.62	10000	300	3.0	0.0	0.0	00:03
7	ALU 6 ENDMILL BOSS RF	T11	6.0	0	9.62	10000	500	6.0	0.1	0.1	00:06
8	ALU 6 ENDMILL BOSS FINISH	T11	6.0	0	9.62	10000	500	6.0	0.0	0.0	00:04
9	ALU 6.0 SLOT MILL SLOT	T13	6.0	0	9.4	10000	300	6.0	--	--	00:10

ESTIMATED SETUP TIME: 01:40 (includes tool changes)

SETUP 5 -- -X approach (left side face)

PRINCIPAL | 3 clusters | 6 operations

Fixture	90° angle plate or tombstone fixture.
Description	Left-side face setup. Rotate part 90° so left face is accessible. Clamp on bottom and right faces.
Rotation	None -- standard orientation
WCS	G58
Origin X/Y/Z	X=CENTER (Y=-20.711mm) Y=CENTER (Z=2.750mm) Z=TOP
Face (W x H)	104.8 x 21.5 mm
Max depth	55.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T07	4.0	0	2.0	10000	600	2.0	--	--	00:01
2	ALU 5 DRILL	T10	5.0	0	1.48	10000	650	1.477	--	--	00:01
3	ALU 3 ENDMILL BOSS RF	T06	3.0	0	3.0	10000	300	3.0	0.1	0.1	00:03
4	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	3.0	10000	300	3.0	0.0	0.0	00:03
5	ALU 3 ENDMILL BOSS RF	T06	3.0	0	4.75	10000	300	3.0	0.1	0.1	00:05
6	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	4.75	10000	300	3.0	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:42 (includes tool changes)

SETUP 6 -- +X approach (right side face)

PRINCIPAL | 2 clusters | 4 operations

Fixture	90° angle plate or tombstone fixture.
Description	Right-side face setup. Rotate part 90° so right face is accessible. Clamp on bottom and left faces.
Rotation	None -- standard orientation
WCS	G59
Origin X/Y/Z	X=CENTER (Y=-20.711mm) Y=CENTER (Z=2.750mm) Z=TOP
Face (W x H)	104.8 x 21.5 mm
Max depth	55.3 mm

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	ALU CENTER DRILL	T02	2.0	0	1.0	10000	300	1.0	--	--	00:01
2	ALU 3 DRILL	T05	3.0	0	7.15	10000	450	7.146	--	--	00:01
3	ALU 3 ENDMILL BOSS RF	T06	3.0	0	3.37	10000	300	3.0	0.1	0.1	00:05
4	ALU 3 ENDMILL BOSS FINISH	T06	3.0	0	3.37	10000	300	3.0	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:35 (includes tool changes)

SETUP 7 -- angled approach [0.64, -0.768, -0.0]

ANGLED | 1 clusters | 3 operations

Fixture	Sine plate or angled fixture set to 140.19°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 140.19° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feat
Rotation required	140.19 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [-0.64, 0.768, 0.0] is not aligned to a principal axis -- part must be rotated or an angled fixture used before drilling**										
2	ALU 1 ENDMILL BOSS ANGLED RF	T01	1.0	0	0.05	10000	100	0.051	0.1	0.1	00:03
3	ALU 1 ENDMILL BOSS ANGLED FINISH	T01	1.0	0	0.05	10000	100	0.051	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:15 (includes tool changes)

SETUP 8 -- angled approach [-0.707, -0.707, -0.0]

ANGLED | 1 clusters | 3 operations

Fixture	Sine plate or angled fixture set to 135.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 135.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All featur
Rotation required	135.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [0.707, 0.707, 0.0] is not aligned to a principal axis — part must be rotated or an angled fixture used before drilling**	--	--	--	--	--	--	--	--	--	--
2	ALU 1 ENDMILL BOSS ANGLED RF	T01	1.0	0	0.49	10000	100	0.491	0.1	0.1	00:03
3	ALU 1 ENDMILL BOSS ANGLED FINISH	T01	1.0	0	0.49	10000	100	0.491	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:15 (includes tool changes)

SETUP 9 -- angled approach [0.707, -0.707, -0.0]

ANGLED | 1 clusters | 3 operations

Fixture	Sine plate or angled fixture set to 135.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 135.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All featu
Rotation required	135.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [-0.707, 0.707, 0.0] is not aligned to a principal axis — part must be rotated or an angled fixture used before drilling**										
2	ALU 1 ENDMILL BOSS ANGLED RF	T01	1.0	0	0.49	10000	100	0.491	0.1	0.1	00:03
3	ALU 1 ENDMILL BOSS ANGLED FINISH	T01	1.0	0	0.49	10000	100	0.491	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:15 (includes tool changes)

SETUP 10 -- angled approach [0.707, 0.707, -0.0]

ANGLED | 1 clusters | 3 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [-0.707, -0.707, 0.0] is not aligned to a principal axis -- part must be rotated or an angled fixture used before drilling**	-	-	-	-	-	-	-	-	-	-
2	ALU 1 ENDMILL BOSS ANGLED RF	T01	1.0	0	0.49	10000	100	0.491	0.1	0.1	00:03
3	ALU 1 ENDMILL BOSS ANGLED FINISH	T01	1.0	0	0.49	10000	100	0.491	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:15 (includes tool changes)

SETUP 11 -- angled approach [-0.707, 0.707, -0.0]

ANGLED | 1 clusters | 3 operations

Fixture	Sine plate or angled fixture set to 45.0°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 45.0° around -Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All feature
Rotation required	45.0 deg around -Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [0.707, -0.707, 0.0] is not aligned to a principal axis — part must be rotated or an angled fixture used before drilling**	--	--	--	--	--	--	--	--	--	--
2	ALU 1 ENDMILL BOSS ANGLED RF	T01	1.0	0	0.49	10000	100	0.491	0.1	0.1	00:03
3	ALU 1 ENDMILL BOSS ANGLED FINISH	T01	1.0	0	0.49	10000	100	0.491	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:15 (includes tool changes)

SETUP 12 -- angled approach [0.64, 0.768, -0.0]

ANGLED | 1 clusters | 3 operations

Fixture	Sine plate or angled fixture set to 39.81°, or 4th/5th axis rotary table if available.
Description	Angled setup. Tilt part 39.81° around +Z axis (B-axis tilt) from the default orientation. Verify with dial indicator before machining. All featu
Rotation required	39.81 deg around +Z axis (B-axis tilt) -- verify with dial indicator before cutting
WCS	G59
Origin X/Y/Z	X=CENTER Y=CENTER Z=TOP
Face (W x H)	--
Max depth	--

OPERATIONS

No.	ToolPath Name	T#	Dia (mm)	Tool R (mm)	Depth MAX	RPM	Feed (mm/min)	DOC (mm)	Stk XY (mm)	Stk Z (mm)	Time
1	**FIXTURE ROTATION -- Feature axis [-0.64, -0.768, 0.0] is not aligned to a principal axis — part must be rotated or an angled fixture used before drilling**										
2	ALU 1 ENDMILL BOSS ANGLED RF	T01	1.0	0	0.05	10000	100	0.051	0.1	0.1	00:03
3	ALU 1 ENDMILL BOSS ANGLED FINISH	T01	1.0	0	0.05	10000	100	0.051	0.0	0.0	00:03

ESTIMATED SETUP TIME: 00:15 (includes tool changes)

WARNINGS & FLAGGED ITEMS

Type	Cluster	Operation	Detail
RPM CAPPED	C0	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C0	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C0	twist_drill	Effective Vc = 78.5 m/min (reduced from 130 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C0	counterbore_mill	Effective Vc = 251.3 m/min (reduced from 420 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C0	counterbore_mill	Effective Vc = 251.3 m/min (reduced from 420 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C1	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C1	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C1	twist_drill	Effective Vc = 78.5 m/min (reduced from 130 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C1	counterbore_mill	Effective Vc = 251.3 m/min (reduced from 420 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C1	counterbore_mill	Effective Vc = 251.3 m/min (reduced from 420 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C2	spot_drill	Effective Vc = 125.7 m/min (reduced from 250 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C2	twist_drill	Effective Vc = 157.1 m/min (reduced from 160 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C3	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C3	twist_drill	Effective Vc = 94.2 m/min (reduced from 140 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C4	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C4	twist_drill	Effective Vc = 78.5 m/min (reduced from 130 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C4	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C5	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C5	twist_drill	Effective Vc = 78.5 m/min (reduced from 130 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C5	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C6	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C6	twist_drill	Effective Vc = 78.5 m/min (reduced from 130 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C6	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C7	twist_drill	SUBSTITUTION: 2.529mm is non-standard. Using nearest standard size 2.5mm. Verify hole tolerance before confirming.
RPM CAPPED	C7	twist_drill	Effective Vc = 78.5 m/min (reduced from 130 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.

Type	Cluster	Operation	Detail
RPM CAPPED	C7	counterbore_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C7	counterbore_mill	Effective Vc = 125.7 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C8	circular_interp	Effective Vc = 251.3 m/min (reduced from 420 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C10	spot_drill	Effective Vc = 62.8 m/min (reduced from 80 m/min). Feed rate adjusted accordingly.
SUBSTITUTION	C10	twist_drill	SUBSTITUTION: 3.2mm is non-standard. Using nearest standard size 3.0mm. Verify hole tolerance before confirming.
RPM CAPPED	C10	twist_drill	Effective Vc = 94.2 m/min (reduced from 140 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C11	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C11	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C12	contour_mill	Effective Vc = 62.8 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C12	contour_mill	Effective Vc = 62.8 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C13	contour_mill	Effective Vc = 62.8 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C13	contour_mill	Effective Vc = 62.8 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C14	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C14	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C15	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C15	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C16	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C16	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C17	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C17	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C18	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C18	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C19	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C19	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C20	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C20	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C21	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C21	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C22	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C22	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C23	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C23	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C24	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C24	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C25	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C25	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C26	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C26	contour_mill	Effective Vc = 31.4 m/min (reduced from 200 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C27	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.

Type	Cluster	Operation	Detail
RPM CAPPED	C27	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C28	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C28	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C29	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C29	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C30	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C30	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C31	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C31	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C32	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C32	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C33	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C33	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C34	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C34	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C35	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C35	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C36	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C36	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C37	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C38	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C38	contour_mill	Effective Vc = 94.2 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C39	contour_mill	Effective Vc = 188.5 m/min (reduced from 400 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C39	contour_mill	Effective Vc = 188.5 m/min (reduced from 400 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C40	contour_mill	Effective Vc = 188.5 m/min (reduced from 400 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C40	contour_mill	Effective Vc = 188.5 m/min (reduced from 400 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C83	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C84	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C85	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C86	chamfer_mill	Effective Vc = 188.5 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C88	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C89	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C90	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C91	slot_mill	Effective Vc = 251.3 m/min (reduced from 380 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C92	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C93	slot_mill	Effective Vc = 188.5 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C94	slot_mill	Effective Vc = 188.5 m/min (reduced from 350 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C95	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C96	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.

Type	Cluster	Operation	Detail
RPM CAPPED	C97	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C98	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.
RPM CAPPED	C99	slot_mill	Effective Vc = 125.7 m/min (reduced from 300 m/min). Feed rate adjusted accordingly.